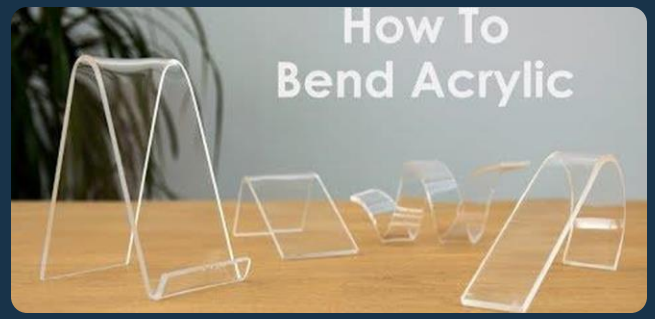
Rail Design Options

Three candidates for the UAV launch rail



Clear Acrylic Sheets

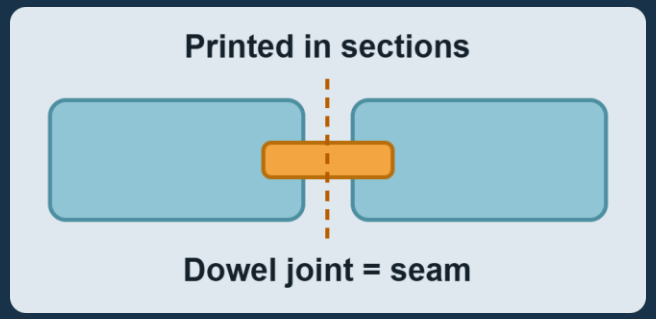
Laser-cut to size with drain holes, then heat-bent into one continuous rail.

COST

$98–$138 per sheet, plus laser cutting

TENSILE STRENGTH

65 MPa



3D Printed PLA

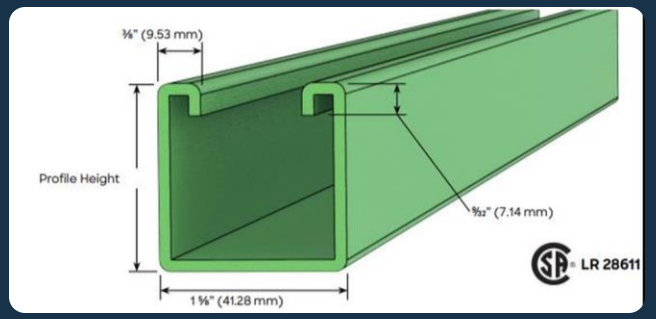
Printed in multiple sections joined by dowels; drain holes drilled afterwards.

COST

$90–$120 for 3–4 filament rolls

TENSILE STRENGTH

50–60 MPa



Atkore Sasco Strut

Slotted S2 galvanized steel channel cut to length, paired with an S875T-EG trolley.

COST

~$0, surplus from a team member's work

TENSILE STRENGTH

400–841 MPa (galvanized)

Pros and Cons

OPTION 1

Molded Clear Acrylic

PROS

- One continuous piece
- Smooth, low-friction surface
- Easy to laser-cut drain holes

CONS

- Difficult to bend properly
- High likelihood of fabrication error
- Costly when errors repeat

OPTION 2

3D Printed PLA

PROS

- Fully customizable shape
- Cheaper than acrylic
- Easy to fabricate

CONS

- Seams are weak points and catch hazards
- Weakest option, even at 100% fill
- Higher friction than acrylic

OPTION 3 · SELECTED

Atkore Sasco Strut

PROS

- Most structurally sound
- Only cutting required; trolleys made to fit
- Pre-slotted for drainage
- ~$0 cost

CONS

- Heavier, more load on actuators
- Galvanized steel corrodes over time
- Higher friction than acrylic

FINAL CHOICE

Atkore Sasco Strut

S2 SL2-G 120 channel with a matching S875T-EG trolley

Most stable

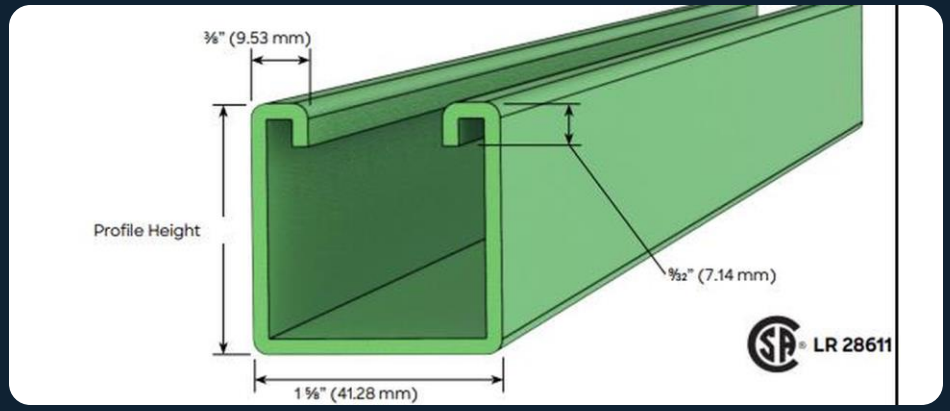
Solid steel gives the most consistent launches.

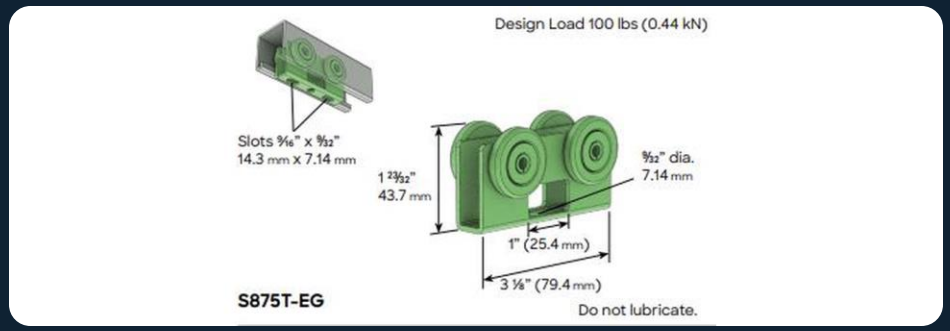
Least fabrication

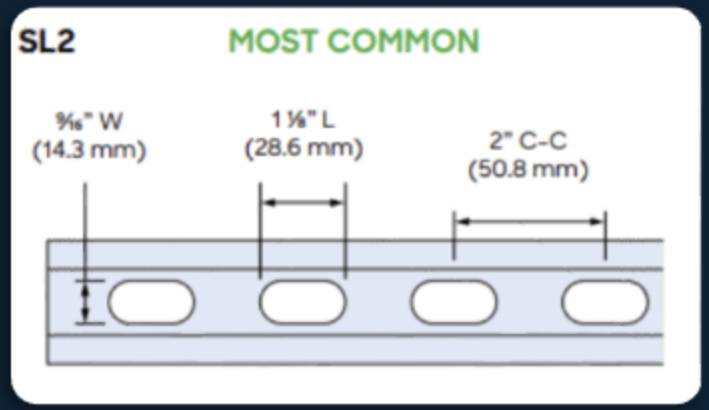
One solid piece with drainage holes, just cut to length.

Trolley

The trolley is made for this strut and is structurally sound.

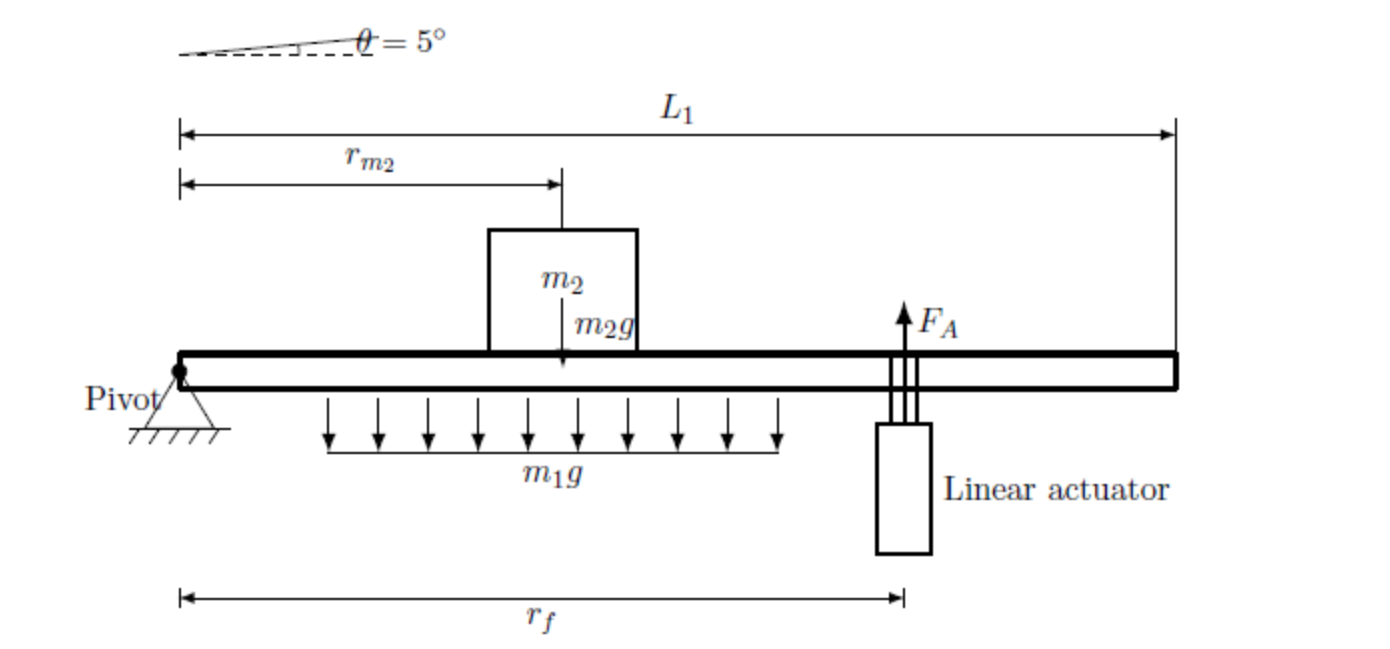






Trade-offs accepted: added weight on the angle control actuators; galvanized steel will corrode over time.

Front Rail Linear Actuation Requirments



Free Body Diagram on rail system

Final Equation Place holders

$$F_A = \frac{m_1 g r_{m1} + m_2 g r_{m2} + I_T(\frac{2\Delta\theta}{t^2})}{r_f}$$

Linear Force Req

$$\Delta z = L_1 sin\theta$$

Linear Distanced Req

$$\frac{dz}{dt} = \frac{\Delta z}{t}$$

Linear Speed Req

Temp values for early comparison

- L_1 = 1.2 m
- r_f = 1 m
- r_{m2} = 0.25 m
- m_2 = 3kg (plane 6kg total)
- $\Delta\theta$ = 5°
- T = 0.5 s
- m_2 = varying per rail type

Sasco Strut
- F_A= 25.778 N, 50N SF

3d Printed PLA
- F_A= 14.95 N, 30N SF

Sasco Strut
- F_A= 14.65 N, 30N SF

Δz = 0.0873 m or 90 mm

$\frac{dz}{dt}$ = 0.1743 m/s or 170mm/s

Possible Linear Actuators



Lead Screw

Fig. 7. Integrated motor and screw design concept. Source: Linear Motion Tips, "What options are there for integrated motor and screw designs?"



Ball Screw

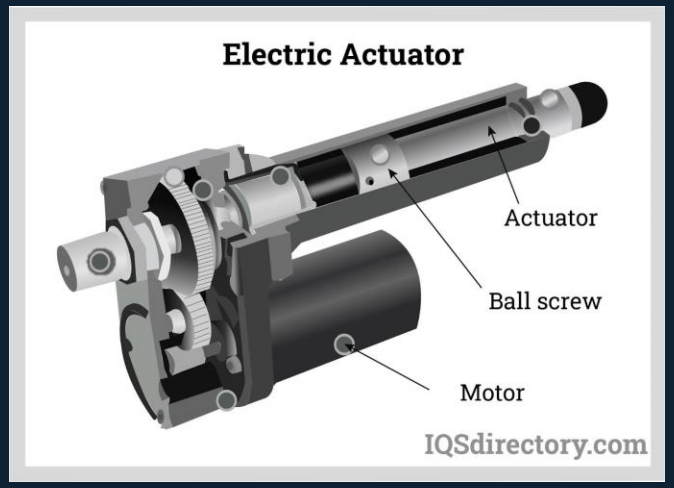
Fig. 8. Ball screw vs. lead screw comparison. Source: Rockford Ball Screw, "Ball Screw vs Lead Screw: Everything You Need to Know," Jan. 25, 2023.

Possible Linear Actuators Cont.



Linear Motor

Fig. 9. Operating principle of a linear motor. Source: ITG Motor, "What Is a Linear Motor?"



Pre-Built Electric Actuator

Fig. 10. Types of linear actuators. Source: IQS Directory, "Types of Linear Actuators."



Servo motor at hinge

Fig. 11. NEMA 23 integrated servo motor used in the actuator design. Source: STEPPERONLINE, "NEMA 23 Integrated Easy Servo Motor."